📘 EXAM FORMULA SHEET

BEEE

Basic Electrical & Electronics Engineering

⚡ UNIT III

Energy Resources · Electricity Billing · Safety Measures

B.Tech 1st Year | 1st Semester | Last-Day Revision Tool

All Formulas · Concepts · Tables · Flashcards · Exam Tips

Hydel Power · Thermal Power · Nuclear Power · Solar Energy · Wind Energy · Electricity Billing · Fuse & MCB · Earthing · Electric Shock · Safety Measures

⚡ KEY FORMULAS + ENERGY RESOURCES OVERVIEW

📐 MASTER FORMULA REFERENCE

ELECTRICAL POWER

$P = V \times I$

P=Watts, V=Volts, I=Amps

ENERGY CONSUMED

$E = P \times t$

E=Wh, P=Watts, t=hours

ENERGY IN KWH (UNITS)

$\text{Units} = (P \times t)/1000$

1 Unit = 1 kWh = 1000 Wh

EFFICIENCY

$\eta = (\text{Output}/\text{Input}) \times 100\%$

η = eta (Greek)

ELECTRICITY BILL COST

$\text{Cost} = \text{Units} \times \text{Tariff}$

Tariff in ₹/unit

OHM'S LAW

$V = I \times R$

V=Volts, I=Amps, R=Ohms

💡 **Remember:** 1 kWh = 1 Unit of electricity = 1000 Watts consumed for 1 hour = 3.6×10^6 Joules

🔋 ENERGY RESOURCES – CLASSIFICATION

Aspect	Conventional (Non-Renewable)	Non-Conventional (Renewable)
Sources	Thermal, Hydel, Nuclear	Solar, Wind, Tidal, Geothermal
Availability	Limited / Exhaustible	Unlimited / Inexhaustible
Pollution	High (Thermal/Nuclear)	Low / Zero
Cost	High running cost	High setup, low running cost
Reliability	High, 24×7 supply	Depends on weather/nature
Examples	Coal, Oil, Gas, Uranium, Water	Sunlight, Wind, Biomass

🌊 HYDEL (HYDROELECTRIC) POWER PLANT

Layout Components (in order)

Dam → Reservoir → Penstock → Turbine → Generator → Transformer → Grid

Working Principle

- Water stored in reservoir (potential energy)
- Water flows through penstock (kinetic energy)
- Turbine rotates due to water force
- Generator converts mechanical → electrical energy

Energy Conversion Chain

Potential Energy → Kinetic Energy → Mechanical Energy → Electrical Energy

✅ Advantages

- No fuel cost, renewable source
- No air/water pollution
- Long life (50+ years)
- Can respond quickly to load changes
- Flood control & irrigation benefits

❌ Disadvantages

- High initial cost
- Dependent on rainfall/season
- Displaces population (large dams)
- Ecological disturbance
- Silting of reservoir over time

🔥 THERMAL · ☢️ NUCLEAR · 🌞 SOLAR · 💨 WIND POWER

🔥 THERMAL POWER PLANT

Working Principle

Fuel (Coal) → Combustion → Steam → Turbine → Generator

- Coal burned in boiler → high-pressure steam
- Steam drives turbine (mechanical energy)
- Generator converts to electrical energy
- Condenser converts exhaust steam → water (recycled)

Key Components

- Boiler, Turbine, Generator, Condenser, Cooling Tower

Limitations: Non-renewable fuel, air pollution (CO_2, SO_2), high running cost, slow start-up

☢️ NUCLEAR POWER PLANT

Working Principle

Nuclear Fission → Heat → Steam → Turbine → Generator

- Uranium-235 / Plutonium-239 undergoes fission
- Releases enormous heat energy
- Heat converts water to steam
- Steam drives turbine → generator produces electricity

Key Components

- **Reactor:** Where fission occurs
- **Moderator:** Slows neutrons (heavy water, graphite)
- **Control Rods:** Absorb excess neutrons (Boron, Cadmium)
- **Coolant:** Carries heat away (water, CO_2)
- **Shielding:** Prevents radiation leakage

Aspect	🌊 Hydel	🔥 Thermal	☢️ Nuclear
Fuel	Water	Coal/Oil/Gas	Uranium/Plutonium
Pollution	None	High (CO_2, SO_2)	Radiation risk
Running Cost	Very Low	High	Low
Setup Cost	High	Medium	Very High
Fuel Availability	Renewable	Exhaustible	Limited
Output	Depends on water	Consistent	Very High

🌞 SOLAR POWER GENERATION

Types

- **PV (Photovoltaic) Cells:** Direct sunlight → DC electricity
- **Solar Thermal:** Sunlight → Heat → Steam → Electricity

Working Principle

Photoelectric Effect: Photons from sunlight strike semiconductor (Silicon) → free electrons → electric current (DC)

✅ Advantages

- Free fuel (sunlight), zero emissions
- Low maintenance, silent operation
- Suitable for remote areas
- Long lifespan (~25 years)

❌ Disadvantages

- High initial cost
- Works only during daylight
- Efficiency depends on weather/location
- Large area required

💨 WIND POWER GENERATION

Working Principle

Kinetic Energy of Wind → Turbine Blades Rotate → Generator → Electricity

Key Components

- Wind turbine blades (aerofoil shape)
- Rotor, Gearbox, Generator
- Nacelle (houses mechanical parts)
- Tower, Foundation

Wind Power Formula

POWER IN WIND

$$P = \frac{1}{2}\rho A v^3$$

ρ=air density, A=swept area, v=wind speed

✅ Advantages

- Free fuel, renewable, no emissions
- Low running cost
- Land below turbines usable (farming)

❌ Disadvantages

- Depends on wind speed/direction
- Noise, visual impact, bird hazard
- Suitable only in wind-prone areas

💡 ELECTRICITY BILLING – COMPLETE GUIDE

KEY DEFINITIONS & CONCEPTS

1 Unit of Electricity = 1 kWh
= Power of 1000 W (1 kW) used for 1 hour
= 3.6×10^6 Joules = 3600 kJ

ENERGY IN UNITS (KWH)

$$E\ (kWh) = [P(W) \times t(hr)] / 1000$$

P = Power in Watts, t = Time in hours

TOTAL ELECTRICITY BILL

Bill = Fixed Charge + (Units × Rate)

Two-Part Tariff System

Two-Part Tariff:
- Fixed Charge = ₹/month (based on connected load)
- Running/Variable Charge = ₹/unit consumed

COMMON APPLIANCE POWER RATINGS

Appliance	Power Rating	Typical Daily Use	Units/Day
Ceiling Fan	60–75 W	8–10 hrs	0.6 units
LED Bulb	7–15 W	8 hrs	0.1 units
CFL Bulb	15–23 W	8 hrs	0.16 units
Tubelight	36–40 W	8 hrs	0.3 units
Refrigerator	150–250 W	24 hrs	3.0–4.0 units
Air Conditioner (1.5 ton)	1500–2000 W	8 hrs	12–16 units
Washing Machine	500–1000 W	1 hr	0.5–1.0 units
Electric Iron	750–1000 W	1 hr	0.75–1.0 units
Laptop Computer	50–100 W	8 hrs	0.5–0.8 units
Desktop Computer	200–400 W	8 hrs	1.6–3.2 units
Printer (inkjet)	15–30 W	1 hr	0.03 units
Microwave Oven	800–1200 W	0.5 hr	0.4–0.6 units
Water Heater (Geyser)	2000 W	0.5 hr	1.0 unit
Electric Heater (Room)	1000–2000 W	4 hrs	4.0–8.0 units
Television (LED 32")	30–80 W	6 hrs	0.2–0.5 units

ELECTRICITY BILL CALCULATION – STEP BY STEP

Method

Step 1: Find Power of each appliance (W)
Step 2: Find daily usage time (hours)
Step 3: Energy/day = (P × t) / 1000 kWh
Step 4: Monthly energy = Energy/day × 30
Step 5: Bill = Fixed charge + (Total units × Tariff)

Sample Problem

Q: A 1500W AC runs 6 hrs/day for 30 days. Tariff = ₹5/unit. Fixed = ₹50. Find bill.

Sol:
Energy = (1500 × 6 × 30) / 1000 = **270 kWh**
Bill = ₹50 + (270 × 5) = ₹50 + ₹1350 = **₹1400**

⚙️ SAFETY MEASURES – FUSE · MCB · EARTHING

FUSE

Definition

A thin wire of low melting point that melts and breaks the circuit when excess current flows, protecting the circuit.

Working Principle

Heating Effect of Current (I^2R) – When current exceeds rated value, fuse wire heats up and melts → circuit opens → equipment protected.

Formula

HEAT GENERATED IN FUSE

$$H = I^2 \times R \times t$$

H=Joules, I=Amps, R=Ohms, t=seconds

Characteristics

- Made of tin-lead alloy or copper
- Low melting point (< 200°C)
- Fusing factor = Min fusing current / Current rating
- Must be replaced after operation (one-time use)

MCB (Miniature Circuit Breaker)

Definition

An automatic switching device that interrupts current during overload or short circuit and can be reset without replacement.

Working Principle

- **Overload (Thermal):** Bimetallic strip bends due to heat → trips MCB
- **Short Circuit (Magnetic):** Electromagnetic coil produces force → trips MCB instantly

Key Feature: Automatically trips and can be manually RESET – no replacement needed!

FUSE vs MCB – COMPARISON

Feature	Fuse	MCB
Working Principle	Heating effect of current	Thermal + Electromagnetic
Operation	Melts & breaks circuit	Trips (switches off)
Reset	Must be replaced	Can be reset (reusable)
Speed of Operation	Slow (thermal)	Fast (especially for short circuit)
Cost	Low (element)	High (device)
Indication	No visual indication	Clear trip indication
Safety	Moderate	Higher
Protects against	Overload & short circuit	Overload & short circuit
Installation	Simple	More complex
Use	Old buildings, low-cost	Modern installations

EARTHING (GROUNDING)

Purpose

To provide a safe path for fault current to flow into the earth, preventing electric shock to persons touching electrical equipment with a fault.

Type 1: Plate Earthing

- A copper/GI plate buried ~3m deep in earth
- Plate size: 60cm × 60cm × 3.15mm (copper)
- Surrounded by alternate layers of salt and charcoal
- Used for: Substations, industrial areas
- More reliable, suitable for high fault currents

Type 2: Pipe Earthing

- A GI pipe buried vertically in ground
- Pipe: 38mm diameter, 2m length (minimum)
- Surrounded by charcoal and salt for low resistance
- Used for: Domestic/residential wiring
- Most common method, easy installation

Feature	Plate Earthing	Pipe Earthing
Electrode	Metal plate (Cu/GI)	GI pipe

Depth	~3 meters	~2 meters (min)
Area Required	More space	Less space
Maintenance	More maintenance	Easy to maintain
Usage	Substations, heavy industry	Homes, offices
Cost	Higher	Lower

⚠️ ELECTRIC SHOCK & SAFETY PRECAUTIONS

⚡ ELECTRIC SHOCK

Definition

Electric shock is the physiological reaction of the human body when electric current passes through it.

Common Causes

- Touching live (Phase) wire directly
- Faulty insulation on equipment
- Absence of earthing
- Working on live circuits without precaution
- Using damaged electrical appliances
- Wet hands touching electrical equipment
- Children inserting objects in sockets

Effects on Human Body

Current Level	Effect
1 mA	Tingling sensation
5 mA	Slight shock, not painful
10–20 mA	Cannot let go (muscular lock)
50 mA	Very painful, severe effects
100–300 mA	Ventricular fibrillation (fatal)
>300 mA	Heart stops, severe burns

Prevention

- Use earthed equipment
- Use MCBs and RCCBs (Earth Leakage protection)
- Never touch live wire with bare hands
- Use rubber gloves, insulated tools
- Switch OFF before working on circuits

🛡️ SAFETY PRECAUTIONS & MEASURES

✅ DO's (Safe Practices)

- Always switch OFF before repair/maintenance
- Use ISI-marked electrical equipment
- Keep electrical panels/boards dry
- Use proper earthing for all appliances
- Install MCBs/fuses of correct rating
- Use rubber-soled shoes near electrical work
- Use insulated tools (rubber/plastic handles)
- Check wiring periodically for damage
- Install earthing and RCCB for safety
- Report any sparks or burning smells

❌ DON'Ts (Unsafe Practices)

- Never touch live wires with bare hands
- Do not use wet hands near electrical switches
- Never overload sockets with multiple devices
- Do not use damaged/frayed wires
- Never bypass fuses or MCBs
- Do not repair energized (live) circuits
- Do not use higher-rated fuse than recommended
- Avoid loose connections (sparking risk)
- Do not leave appliances on unattended
- Never use electric appliances near water

🔧 ADDITIONAL PROTECTION DEVICES

Device	Full Form	Function	Trips When
MCB	Miniature Circuit Breaker	Overload & Short Circuit protection	Excess current flows
MCCB	Molded Case Circuit Breaker	Higher current protection (industries)	Heavy fault current
RCCB / ELCB	Residual Current CB / Earth Leakage CB	Detects earth leakage (shock prevention)	Current imbalance ≥ 30mA
Fuse	—	Sacrificial protection element	Rated current exceeded
Surge Arrester	—	Protects from voltage surges/lightning	Overvoltage occurs

⚠️ **First Aid for Electric Shock:** (1) Do NOT touch the victim directly. (2) Switch OFF supply immediately or use dry wood/rubber to separate. (3) Call emergency services. (4) Perform CPR if victim is unconscious/not breathing. (5) Keep victim warm until help arrives.

🧠 FLASHCARDS – QUICK MEMORY BOOSTER (30 Cards)

Q1: What is 1 unit of electricity?
A: 1 kWh (kilowatt-hour)

Q2: Formula for electrical energy?
A: $E = P \times t$

Q3: Formula for electrical power?
A: $P = V \times I$

Q4: What is efficiency (η)?
A: $\eta = (\text{Output}/\text{Input}) \times 100\%$

Q5: Formula for electricity bill?
A: Bill = Fixed Charge + (Units × Tariff)

Q6: Working principle of fuse?
A: Heating effect of current (I^2Rt)

Q7: What is earthing?
A: Connecting equipment body to earth for safety

Q8: What is Two-Part Tariff?
A: Fixed Charge + Running (Variable) Charge

Q9: Energy conversion in Hydel Plant?
A: Potential → Kinetic → Mechanical → Electrical

Q10: Working principle of solar PV cell?
A: Photoelectric effect (photons → electrons)

Q11: Working principle of Nuclear plant?
A: Nuclear Fission (U-235 splits → heat → electricity)

Q12: What does MCB stand for?
A: Miniature Circuit Breaker

Q13: Advantage of MCB over Fuse?
A: Reusable – can be reset; no replacement needed

Q14: What is the function of a Moderator in nuclear plant?
A: Slows down neutrons to sustain fission

Q15: What are Control Rods made of?
A: Boron or Cadmium (absorb excess neutrons)

Q16: 1 kWh = ? Joules?
A: 3.6×10^6 Joules (3,600,000 J)

Q17: What is the function of Penstock?
A: Carries water from reservoir to turbine (Hydel plant)

Q18: What fatal current level causes ventricular fibrillation?
A: 100–300 mA

Q19: What causes electric shock?
A: Electric current passing through human body

Q20: Most common earthing method for homes?
A: Pipe Earthing

Q21: What is RCCB?
A: Residual Current Circuit Breaker – detects earth leakage

Q22: Power consumption of a 1.5 ton AC?
A: 1500–2000 Watts

Q23: What is the wind power formula?
A: $P = \frac{1}{2}\rho A v^3$

Q24: What is fusing factor?
A: Min fusing current ÷ Current rating (always >1)

Q25: Non-conventional energy examples?
A: Solar, Wind, Tidal, Geothermal, Biomass

Q26: What material is used in solar cells?
A: Silicon (semiconductor)

Q27: Ohm's Law formula?
A: $V = I \times R$

Q28: First action in case of electric shock?
A: Switch OFF the power supply immediately

Q29: What is the role of coolant in nuclear plant?
A: Carries heat from reactor to steam generator

Q30: What overload protection does MCB use?
A: Bimetallic strip (thermal) for overload; EM coil for short circuit

🚀 EXAM TIPS & COMMON MISTAKES TO AVOID

⚠️ TOP COMMON MISTAKES IN EXAMS

1. **Power vs Energy Units Confusion:** Power is in Watts (W) or kW. Energy is in Watt-hours (Wh) or kilowatt-hours (kWh). NEVER write energy in Watts — Watts is a RATE, not energy. Always include units in answers!

2. **Forgetting 1 Unit = 1 kWh:** In bill calculations, energy must always be converted to kWh (Units). Divide by 1000 when converting Wh → kWh. Common error: multiplying directly without dividing by 1000.

3. **Electricity Bill Calculation Errors:** Always add the Fixed Charge (₹/month) to the variable charge. Many students forget the fixed component. Two-part tariff = Fixed + Running.

4. **Mixing Fuse and MCB Working Principles:** Fuse works on *heating effect of current*. MCB uses *both thermal (bimetallic strip) and electromagnetic (coil) effects*. Do NOT write "fuse trips" — fuses MELT and BREAK. MCBs TRIP.

5. **Ignoring Safety Precautions in Theory Answers:** When asked about any power plant or electrical system, always mention safety aspects. Nuclear: radiation shielding. Electrical: earthing, MCB, fuse. This adds marks!

6. **Wrong Appliance Power Ratings:** Remember: AC (1.5 ton) = ~1500W, Geyser = 2000W, Fan = 60–75W, LED bulb = 7–15W. These often appear in numerical problems. Memorize at least 5–6 key ratings.

7. **Missing Units in Numerical Problems:** Always write: Energy in **kWh**, Power in **W or kW**, Current in **A**, Voltage in **V**, Bill in **₹**. Missing units = lost marks.

8. **Confusing Plate and Pipe Earthing Applications:** Pipe earthing → homes/residential. Plate earthing → substations/heavy industry. Do not reverse them in answers.

9. **Solar vs Thermal Power Plants:** Solar thermal ≠ PV cells. PV cell = direct sunlight to DC electricity (photoelectric effect). Solar thermal = concentrates heat to produce steam. They are different technologies!

10. **Efficiency Formula Errors:** η = (Output/Input) × 100%. Output is always ≤ Input due to losses. If η = 80%, and Input = 500W, then Output = 400W. Common error: switching output and input in formula.

💯 SCORING TIPS FOR UNIT III

For 2-mark questions:
- Definition + Formula / Principle
- Keep it crisp – 2–3 lines max
- Always include units

For 5-mark questions:
- Diagram + Working Principle
- Advantages + Disadvantages
- Applications

For 10-mark questions:
- Full diagram with labels
- Detailed working
- Comparison table
- Formulas with example
- Safety aspects

For Numericals:
- Write GIVEN data first
- Show FORMULA before substitution
- Include UNITS in every step
- Box your final answer

ULTRA QUICK REVISION SHEET – SCAN IN 5 MINUTES

ALL FORMULAS

Power: P = V × I (Watts)

Energy: E = P × t (Wh)

Units (kWh): E = (P×t)/1000

Efficiency: η = (Out/In) × 100%

Ohm's Law: V = I × R

Bill: Fixed + (Units × Rate)

Heat in fuse: $H = I^2Rt$

Wind Power: $P = \frac{1}{2}\rho A v^3$

1 Unit: 1 kWh = 3.6×10^6 J

KEY DEFINITIONS

kWh: Unit of electrical energy consumed

Tariff: Rate charged per unit (₹/kWh)

Fuse: Safety device, melts on overcurrent

MCB: Auto-reset circuit breaker

Earthing: Connecting to ground for safety

Electric Shock: Current through human body

Moderator: Slows neutrons in nuclear reactor

Penstock: Water pipe in hydel plant

Photoelectric Effect: Light → electrons (solar cell)

ENERGY SYSTEMS KEYWORDS

Hydel: Dam, Penstock, Turbine, Generator

Thermal: Coal, Boiler, Steam, Turbine, Condenser

Nuclear: U-235, Fission, Moderator, Control Rods, Coolant, Shielding

Solar PV: Silicon, Photoelectric, DC, Inverter

Wind: Blades, Rotor, Gearbox, Nacelle, Generator

Renewable: Solar, Wind, Hydel

Non-Renewable: Coal, Oil, Gas, Uranium

SAFETY TERMS

MCB: Miniature Circuit Breaker

MCCB: Molded Case Circuit Breaker

RCCB: Residual Current Circuit Breaker

ELCB: Earth Leakage Circuit Breaker

Plate Earthing: GI/Cu plate, 3m deep, industrial

Pipe Earthing: GI pipe, 2m, residential

Fatal current: 100–300 mA (fibrillation)

Fusing factor: Min fusing I ÷ Rated I (>1)

MEMORY AIDS & MNEMONICS

Hydel Energy Conversion: "PKME"
Potential → **K**inetic → **M**echanical → **E**lectrical

Two-Part Tariff: "FR+RR"
Fixed **R**ate + **R**unning **R**ate

Nuclear Plant Components: "MRC COOKS"
Moderator, **R**eactor, **C**ontrol Rods, **C**oolant, **S**hielding

Fuse vs MCB:
Fuse = **MELTS** (one-time use)
MCB = **TRIPS** (reset and reuse)

LAST-MINUTE COMPARISON SNAPSHOT

Power Plant	Fuel	Energy Conversion	Key Component	Pollution
Hydel	Water	PE→KE→ME→EE	Penstock + Turbine	**None**
Thermal	Coal	Chemical→Heat→ME→EE	Boiler + Condenser	**High (CO_2)**
Nuclear	Uranium	Nuclear→Heat→ME→EE	Reactor + Control Rods	**Radiation risk**
Solar PV	Sunlight	Light→DC Electricity	PV Cell (Silicon)	**None**
Wind	Wind	KE→ME→EE	Turbine Blades	**None**